# Secondary Implementation of the OpenSI Smart Gate

**Thea Hari · Janvy Jose · Martin Benny · Antus Jojo · Krishna Sharma**

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## Abstract

Smart Gate uses a camera, an AI processor and a rule set to open a wildlife fence wildlife fence gate for native animals and keep invasive predators out. It has only run has only run on its original hardware, so OpenSI cannot rebuild it from its documentation. We rebuild it on a test bench, measure a baseline and record every record every change so later teams can reproduce it.

## Research question

**How can the existing OpenSI SmartGate be reproduced within reproduced within an independent development environment environment while preserving its core animal-recognition, recognition, simulated gate-control functionality and maintaining traceability of the changes made during redeployment.**

1 WHY IT MATTERS | 2 HOW IT WORKS | 3 HOW WE BUILD IT | 4 HOW WE PROVE SUCCESS

**Seven objectives**

1. Build the prototype from OpenSI's hardware and software
2. Integrate the components into one working system
3. Restore animal detection with the TensorRT model
4. Restore autonomous gate operation
5. Restore manual control from the web interface
6. Keep a chain of custody for every changes
7. Leave a stable base, with no regression

## 1 Background

**75+** native species regularly entangled in barbed wire fences (NSW Government, 2026)

**Fences protect, and also harm.** They keep invasive species out but also block native animals (Sennett & Chambers, 2025).

**Smart Gate is one answer.** It opens for native animals and keeps predators out (AWC, 2025).

**Models have limits.** Rare species and poor conditions reduce accuracy (Goyal et al., 2025), so we measure.



**Figure 1.** Eastern grey kangaroos. Photo: Ricardo Mendonca.

## 2 Value for OpenSI

**A one-off prototype becomes a platform OpenSI can build and test on.**

**BEFORE**
Worked only on its original hardware

→

**AFTER**
**Rebuilt, measured and documented**

**Rebuildable**
Any team can recreate it from our guide

**Trustworthy**
A measured baseline shows nothing got worse

**Traceable**
Every change recorded: what, who, when, why

**Conservation**
Native animals pass; invasive predators stay out

## 3 SmartGate System Architecture: Jetson Orin testbench

1 PIR senses motion → 2 Camera captures → 3 YOLOv5 detects → 4 Ruleset decides → 5 Gate responds

**Web interface:** live video and manual OPEN / CLOSE, independent of detection

**Figure 2.** Signal path. A state machine follows a JSON ruleset.

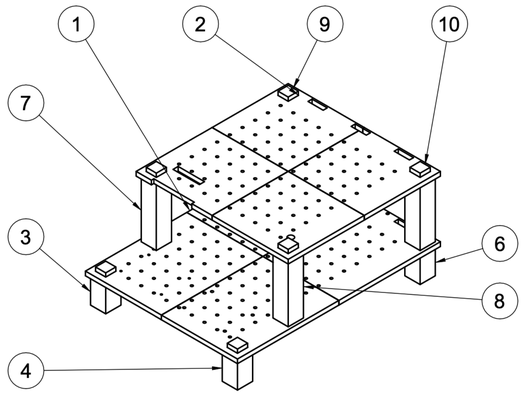
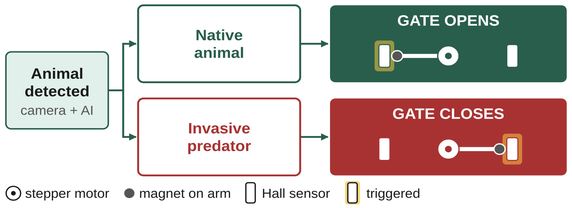



**Figure 3.** CAD design of the 3D-printed test bench

## 4 Baseline and success

We rebuild the original SmartGate on a test bench exactly as it was, then measure it. That working copy is the baseline every later change is compared against.




**Success:** every step works end to end, every time, and another team can rebuild the gate from our guide.

**Figure 4.** Baseline behaviour we reproduce first: native animal, gate opens; invasive predator, gate closes.

## 5 Scope and approach

We rebuild, not redesign. Work is split by role: tasks that can run separately run side by side, and tasks that depend on others wait for them.

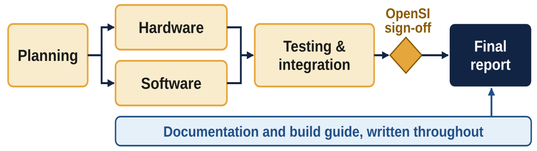



**Figure 5.** Staged rebuild. Each stage is checked before the next; extras start after OpenSI sign-off.

### In scope

- Redeploy on a test bench, with a simulated gate
- Verify detection, autonomous and manual control
- Chain of custody and documentation
- Baseline first, then enhancements

### Out of scope

- A new system from scratch
- Replacing the inherited platform
- Choosing which species may pass
- Field deployment or commercialisation
- Support after handover

## 6 Requirements

| | |
|---|---|
| FR1 | Wake up when motion is sensed |
| FR2 | Capture live camera video |
| FR3 | Move the gate arm with the motor |
| FR4 | Identify the animal |
| FR5 | Decide using the original rules |
| FR6 | Give usable camera images |
| FR7 | Allow safe hardware changes |
| FR8 | Use a suitable power supply |
| FR9 | Allow manual control via the web page |

**Table 1.** Functional requirements. All are Must have and testable as pass or fail.

## 7 Success measures

**Every change must match or beat the baseline.**

| | |
|---|---|
| **Gate performance** | Accuracy, speed and wrong decisions match or beat the baseline |
| **Project delivery** | All 9 requirements pass; no serious faults at handover |
| **OpenSI acceptance** | Written sign-off; next team can continue from our guide |

*Table 2. Success measures. Baseline targets are set once the rebuild is measured.*

## 8 Key Findings and Conclusion

**The original SmartGate vs our rebuild**

| | Original | Our rebuild |
|---|---|---|
| **Computer** | Jetson Nano | **Jetson Orin Nano (newer board)** |
| **AI model** | Only ran on the Jetson Nano | **Same model, rebuilt to run on the Jetson Orin Nano** |
| **False alarms** | A person was mistaken for a deer and shut the gate | **40 of 40 checks with a person in view, no false alarm; still spots foxes (69% sure)** |
| **Simulation of gate operation** | Sliding door that opens and closes | **Indicator arm that swings left or right and stops at a sensor, showing open or closed** |
| **Design** | Every component tucked away in a small, compact box | **Open test bench with visible wiring and accessible components** |
| **Setup instructions** | Not clearly documented | **Wiring diagram, startup guide and build log created** |
| **Website** | Basic manual control page | **Dashboard with live camera feed** |

**Not compared yet:** accuracy, speed and long-running reliability, because the original was never measured.

**Conclusion**

The SmartGate project successfully reconstructed and validated a functional wildlife detection and gate-control test bench, providing OpenSI with a documented foundation for future development and conservation applications.

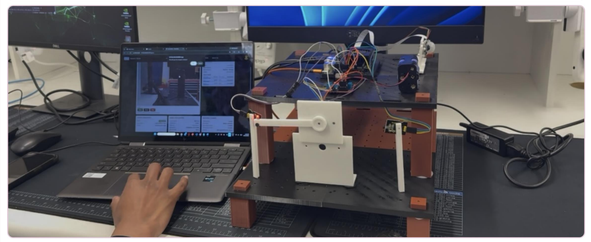


**Figure 6.** The completed test bench, with the live web dashboard on the laptop.

### Key takeaways

- We reproduce an existing system; we do not build a new one.
- A baseline comes first, so no improvement is claimed without evidence.
- Every change is recorded: what, who, when and why.

### Next steps

- Finish the baseline test campaign (Stage I)
- Then fault fixes and any enhancement Alex approves
- Hand over the documented system to OpenSI

### Sources and credits

**Sources cited:**

**Thanks:** OpenSI (Alex and Carlos) and our supervisor, Jeanette.